Sequences 1

A **sequence** is a pattern of numbers or shapes that follow a rule.

2, 4, 6, 8, 10... is a sequence of even numbers.

1, 4, 9, 16, 25... is a sequence of square numbers.

Each number in a number sequence is called a **term**.

You can continue a sequence of numbers by finding the rule to get from one term to the next.

Worked example

The numbers in this sequence increase by the same amount each time:

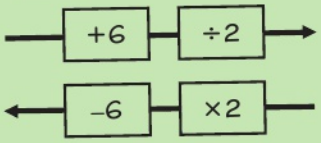
11 15 19 23 27 31

+4 +4

Work out the two missing numbers. **(2 marks)**

Everything in red is part of the answer.

Problem solved! You are given the third term. To work out the term **before** this you need to work **backwards** through the rule. You can use function machines to help you.



Check it!

$(14 + 6) \div 2 = 20 \div 2 = 10$

$(10 + 6) \div 2 = 16 \div 2 = 8$ ✓

Worked example

Here is a sequence. The third term of this sequence is 8.

… … 8 …

The rule for this sequence is

> Add 6 to the previous term and then divide by 2

Work out the first term of this sequence. **(3 marks)**

3rd term = 8

2nd term = $8 \times 2 - 6 = 10$

1st term = $10 \times 2 - 6 = 14$

Generating sequences

You can work out the terms of a sequence by substituting the term number into the *n*th term. Here are some examples:

*n*th term	$9 - 2n$	$n^2 + 10$
1st term	$9 - 2 \times 1 = 7$	$1^2 + 10 = 11$
2nd term	$9 - 2 \times 2 = 5$	$2^2 + 10 = 14$
↓	↓	↓
8th term	$9 - 2 \times 8 = -7$	$8^2 + 10 = 74$

Worked example

The *n*th term of a sequence is $50 - n^2$

Work out the first term of the sequence that is negative. **(2 marks)**

7th term = $50 - 7^2 = 1$

8th term = $50 - 8^2 = -14$

The 8th term is the first negative term.

Now try this

1 The first five terms of a sequence are

3 5 9 15 23

Write down the next two terms of this sequence. **(1 mark)**

2 The *n*th term of a sequence is $7n - 5$

(a) Work out the 8th term of this sequence. **(1 mark)**

(b) Work out the first term of this sequence that is greater than 70. **(1 mark)**

3 The rule for generating a sequence is 'add two consecutive terms to get the next term'.

Here are the first five terms:

2 3 5 8 13

Write down the next three terms of this sequence. **(1 mark)**

This type of sequence is called a **Fibonacci-type** sequence.

Sequences 2

An **arithmetic** or **linear sequence** is a sequence of numbers where the difference between consecutive terms is **constant**. In your exam, you might need to work out the *n*th term of a sequence. Look at this example which shows you how to do it in four steps.

Worked example

1 Here is a sequence:

1 +4 5 +4 9 +4 13 +4 17

Work out a formula for the *n*th term of the sequence. **(2 marks)**

Write in the difference between each term.

2 Here is a sequence:

Zero term

−3 1 +4 5 +4 9 +4 13 +4 17

Work out a formula for the *n*th term of the sequence. **(2 marks)**

Work backwards to find the **zero term** of the sequence. You need to subtract 4 from the first term.

3 Here is a sequence:

Zero term

−3 1 +4 5 +4 9 +4 13 +4 17

Work out a formula for the *n*th term of the sequence. **(2 marks)**

*n*th term = Difference × *n* + Zero term

Write down the formula for the *n*th term. **Remember** this formula for the exam.

4 Here is a sequence:

Zero term

−3 1 +4 5 +4 9 +4 13 +4 17

Work out a formula for the *n*th term of the sequence. **(2 marks)**

*n*th term = Difference × *n* + Zero term

*n*th term = $4n - 3$

Is 99 in this sequence?

You can use the *n*th term to check whether a number is a term in the sequence.

The value of *n* in your *n*th term has to be a **positive** whole number.

Try some different values of *n*:

$n = 25 \rightarrow 4n - 3 = 97$

$n = 26 \rightarrow 4n - 3 = 101$

You can't use a value of *n* between 25 and 26 so 99 is **not** a term in the sequence.

Check it!

Check your answer by substituting values of *n* into your *n*th term.

1st term: when $n = 1$,

$4n - 3 = 4 \times 1 - 3 = 1$ ✓

2nd term: when $n = 2$,

$4n - 3 = 4 \times 2 - 3 = 5$ ✓

You can also generate any term of the sequence.

For the 20th term: when $n = 20$,

$4n - 3 = 4 \times 20 - 3 = 77$

So the 20th term is 77.

Now try this

Here are the first five terms of an arithmetic sequence:

3 7 11 15 19

(a) Write down an expression for the *n*th term. **(2 marks)**

Karen says that 89 is a term in the sequence.

(b) Is she right? Give reasons for your answer. **(2 marks)**

Worked solution video

Write down some terms in the sequence that are close to 89. Remember to write a conclusion to answer the question.